

TECHNICAL PROJECT PLAN

Mixing-Length Convection for HELIOS– VULCAN

*From a closed dry- H_2 column to radiative–convective–photochemical
equilibrium*

Project baseline: bethany-burt/ConvectionMLT + HELIOS/VULCAN + shone/JAX opacity and solver patterns

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Primary destination: rocky exoplanet atmospheres

Secondary destination: sub-Neptunes

First controlled system: dry ideal H_2

Executive summary

RECOMMENDATION Treat the standalone ConvectionMLT repository as a reference laboratory, not as the final HELIOS implementation. Rebuild the core around cell-centred thermodynamics, interface fluxes, a conservative finite-volume energy update, and an interchangeable thermodynamics provider. Validate an explicit dry- H_2 solver first; then port the same flux operator into HELIOS and add faster steady-state solvers only after the reference solution is trusted.

The project should progress through controlled layers of complexity. Each stage adds one major physical or numerical feedback and has a mandatory exit gate. A feature enters the coupled HELIOS–VULCAN model only after it reproduces a limiting case, conserves the relevant quantity, and remains stable under resolution and timestep refinement. This prevents convection, radiation, chemistry, condensation, and solver behaviour from becoming entangled during debugging.

Five design rules govern the programme:

1. Conservation first: compute all energy fluxes on $N+1$ interfaces and update N layers by their conservative divergence.
2. Separate physics from numerics: the explicit timestep is a route to equilibrium; it is not the convective turnover time.
3. Keep numerical, closure, and physical parameters distinct. Grid spacing and timestep are not calibration parameters; the mixing-length coefficient is.
4. Preserve a trusted reference path. The explicit CPU column remains available even after a semi-implicit, Newton, or GPU solver is introduced.
5. Increase coupling strength gradually: convection only → absorption-only radiation → fixed-composition RCE → surfaces and phase change → interactive chemistry/opacity → scattering and clouds.

Phase	Scientific capability	Primary proof
Foundation	Dry H_2 convection in a closed column	Same enthalpy-conserving adiabat for all converged resolutions and non-zero mixing lengths
Radiative foundation	Staggered absorption-only two-stream transfer with fixed opacity	Analytic limits, conservative heating, and Thomas/dense/sweep agreement
Fixed-composition RCE	MLT + radiative flux with production opacity interpolation	Constant total flux, grid-converged RCB, and measured operator bottlenecks
Rocky-planet modules	Multi-component gases, surface, condensation and latent energy	Closed mass and energy budgets before interactive opacity feedback
Coupled thermochemical	VULCAN composition, MLT-derived K_{zz} , gas-opacity feedback, then scattering/clouds	Joint convergence of $T(P)$, composition, opacity, thermodynamics, and flux

How to read this plan

Sections 1–3 define scope, architecture, and the reference equations. Section 4 is the staged roadmap. Sections 5–9 define validation, variables, timescales, performance, and coupling. Section 10 compares the design with existing codes. Sections 11–13 give project governance, immediate actions, and references.

1. Scientific objective and scope

The scientific objective is a one-dimensional atmosphere framework that can calculate radiative–convective equilibrium (RCE) with an explicit mixing-length convective flux, exchange temperature and mixing information with VULCAN, and eventually model high-molecular-weight rocky-planet atmospheres with surfaces and condensable species. The initial H₂ case is a numerical calibration problem, not the final application.

Scope decisions

Decision	Project interpretation
Initial composition	Pure H ₂ represented as a constant-cp ideal diatomic gas. This is a deliberately controlled EOS, not a claim that real H ₂ has constant heat capacity across all temperatures.
Initial convection	Dry Schwarzschild instability and local MLT. No latent heating, condensate loading, or molecular-weight stratification.
Initial geometry	Plane-parallel, hydrostatic, fixed gravity, pressure-coordinate grid.
Initial boundary conditions	Closed convection-only column with zero convective flux at top and bottom.
Use of time	Numerical relaxation to equilibrium. Physical turnover, mixing, radiative, and chemical times are diagnosed separately.
Priority order	Rocky planets first; sub-Neptunes second. H ₂ is the clean validation medium.

Explicit non-goals for the first release

6. No moist adiabat or latent heat in the closed-column milestone.
7. No assumption that MLT alone supplies K_{zz} in radiative regions.
8. No calibration of the mixing-length coefficient against observations before numerical convergence is established.
9. No fully synchronous radiation–chemistry–convection time integration until steady coupled solutions are reliable.

10. No claim that a 1D local MLT closure captures plumes, rotation, shear, or horizontally inhomogeneous convection.

2. Target software architecture

Use one set of physics kernels across the standalone test model and HELIOS. Thermodynamic state and opacity live at layer centres; radiative, convective, latent, and sensible fluxes live at interfaces. The standalone model provides transparent reference solvers; HELIOS remains an integration and comparison target; VULCAN owns chemical kinetics and species transport. The project-owned radiative kernel and opacity adapter must remain independently testable so later improvements do not invalidate the MLT reference.

Component	Owens	Must not own
Grid	Pressure centres/interfaces, layer mass, hydrostatic height	Convection closure or radiative assumptions
Thermodynamics provider	ρ , c_p , μ , ∇_{ad} , enthalpy, optional derivatives	Timestepping and flux divergence
MLT kernel	Instability, mixing length, velocity, F_{conv} , K_{zz_conv}	Profile evolution or plotting
Radiative kernel	Two-stream interface fluxes, boundary conditions, conservative heating	Opacity acquisition or chemistry
Opacity provider	Species tables, P–T interpolation, mixture assembly, spectral metadata	RCE iteration or phase-change bookkeeping
Energy operator	Conservative flux divergence and boundary fluxes	Opacity or reaction networks
Solver	Explicit, semi-implicit, or steady nonlinear update	Physical closure choices
Diagnostics	Budgets, residuals, timescales, convergence and performance	State mutation
Adapters	HELIOS and VULCAN array/layout conversion	Duplicate physics formulas

Recommended repository layout

```

convection_mlt/
  grid.py           pressure grid, layer mass, hydrostatic height
  thermodynamics.py constant-H2 and later general providers
  closure.py        interface MLT flux, velocity, length, Kzz
  radiation.py      absorption-only two-stream reference kernel
  tridiagonal.py    NumPy Thomas reference; JAX production solve
  opacities.py      fixed and shone-backed opacity providers
  energy.py         conservative divergence and boundaries
  solvers.py        explicit reference; later implicit/steady
  diagnostics.py    budgets, residuals, timescales

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adapters/helios.py HELIOS data mapping
 adapters/vulcan.py VULCAN T(P), composition and Kzz mapping
 tests/ unit, invariant, convergence, regression
 experiments/ reproducible parameter sweeps
 legacy/ preserved v2/v3 scripts and notebooks

MIGRATION RULE Do not delete the existing v2/v3 scripts until every useful profile, parameter sweep, and diagnostic has a named replacement. Freeze them under legacy/ and stop extending them once the new reference kernel passes Stage 1.

Grid and state convention

Quantity	Location	Shape
Pressure, temperature, density, cp, mu, composition	Layer centres	N
Pressure interfaces, radiative/convective/latent/sensible flux	Layer interfaces	N+1
Internal MLT velocity, mixing length, Kzz	Internal interfaces; boundary values set explicitly	N+1
Layer mass per unit area	Layer-integrated	N

Use a single sign convention everywhere: positive flux is upward; pressure increases downward. Store the convention in module documentation and assert it in unit tests.

3. Reference physics and conservative update

For adjacent layer centres bracketing interface j , define the atmospheric logarithmic gradient and instability excess as

$$\text{nabla}_j = (\ln T_{\text{lower}} - \ln T_{\text{upper}}) / (\ln P_{\text{lower}} - \ln P_{\text{upper}}); \quad \Delta_j = \max(\text{nabla}_j - \text{nabla}_{\text{ad},j}, 0).$$

For the first closure, follow the simple pressure-coordinate formulation used by AGNI and Lee, Tan & Tsai:

$$H_{p,j} = R_{\text{specific}} T_j / g_j; \quad \ell_j = \alpha H_{p,j}; \quad w_j = \ell_j \sqrt{(g_j / H_{p,j}) \Delta_j}.$$

$$F_{\text{conv},j} = (1/2) \rho_j c_{p,j} w_j T_j (\ell_j / H_{p,j}) \Delta_j; \quad K_{zz,\text{conv},j} = w_j \ell_j.$$

The coefficient $1/2$ and the exact mixing-length convention are closure choices. Record them in model metadata. Do not absorb them silently into α .

For layer i with column mass $\Delta m_i = (P_{\text{bottom}} - P_{\text{top}})/g$, update temperature from total interface flux by

$$dT_i/dt = [F_{tot,bottom(i)} - F_{tot,top(i)}] / [c_{p,i} \Delta m_i], \text{ where } F_{tot} = F_{rad} + F_{conv} + F_{lat} + F_{sens} + F_{other}.$$

This finite-volume form makes internal fluxes telescope. With zero top and bottom flux, the column enthalpy is conserved to solver tolerance. With imposed boundary fluxes, the enthalpy change must equal the time-integrated net boundary input.

Dry-H₂ baseline

$$\mu = 2.01588e-3 \text{ kg mol}^{-1}; \quad R_{specific} = R/\mu; \quad c_p = (7/2) R_{specific}; \quad \text{nabla}_{ad} = R_{specific}/c_p = 2/7.$$

Validity note: this fixed 7/2 heat capacity assumes rotational modes are active and vibrational modes are inactive. Later stages replace it with a temperature- and composition-dependent provider.

Boundary treatment

11. Closed-column validation: $F_{conv} = 0$ at both boundaries.
12. Gas-giant/deep atmosphere: $ell = \alpha H_p$ is an adequate baseline.
13. Rocky atmosphere with a surface: add a surface-aware limiter such as AGNI's Blackadar-like $ell = k_v z / (1 + k_v z/H_p)$, plus a distinct sensible heat flux.
14. Radiative regions: set $F_{conv} = 0$. Any non-zero K_{zz} there must come from a documented overshoot or background-mixing prescription, not from relabelling the MLT result.

Radiative-transfer baseline

Use the same staggering for radiation: T , P , density and opacity are evaluated at N layer centres, while upward, downward and net radiative fluxes are evaluated at $N+1$ interfaces. For mass opacity κ in layer i ,

$$\Delta \tau_{i,\lambda} = \kappa_{i,\lambda} \Delta P_i / g_i; \quad (dT_i/dt)_{rad} = [F_{rad,bottom(i)} - F_{rad,top(i)}] / [c_{p,i} \Delta m_i].$$

The first radiative kernel is two stream, pure absorption and fixed composition. Derive the layered equations in tridiagonal form, solve a small transparent reference with a NumPy Thomas algorithm, and retain an independent directional sweep/dense solve for verification. The production implementation may use JAX compilation, batching and automatic differentiation, but JAX is the execution framework rather than the physical closure or the matrix algorithm.

DEPENDENCY RULE Move interactive composition–opacity feedback later, but do not postpone the radiative operator or fixed-composition opacity provider. Dry-surface and latent-energy tests need a trusted energy-flux framework; they do not need two-way photochemical opacity feedback.

4. Staged development roadmap

Stages are ordered by coupling strength, not by scientific prestige. Every exit gate is required before the next stage becomes the default branch.

0. Foundation and reproducibility

Objective. Create a small, testable package while retaining the existing repository history.

Build. Correct the constant- H_2 heat capacity; adopt centre temperatures/interface fluxes; define units, signs, configuration objects, deterministic outputs, and CI.

Variables to sweep. No broad science sweep. Use two tiny deterministic fixtures and one production-sized case.

15. Unit and dimensional tests for cp , ρ , H_p , $nabla_{ad}$, layer mass, height integration, and array shapes.
16. Static checks that every public result records git commit, configuration, closure prefactor, units, and convergence status.
17. Legacy comparison that reproduces one v2/v3 result closely enough to explain differences from the corrected discretisation.

EXIT GATE The package installs in a clean environment; all deterministic unit tests pass; one command reproduces a baseline result and metadata.

1. Closed dry- H_2 convective column

Objective. Prove that the convection operator recovers the correct enthalpy-conserving adiabat independent of numerical resolution, stable timestep, and non-zero mixing length.

Build. Convection only, fixed gravity, constant cp and μ , zero boundary flux, explicit global timestep, analytic final adiabat.

Variables to sweep. $N = 25, 50, 100, 200, 400$; $\alpha = 0.25, 0.5, 1, 2, 4$; uniform and perturbed log-pressure grids; multiple superadiabatic initial profiles; timestep safety factors across stable and unstable regimes.

18. Stable-profile test: subadiabatic input remains unchanged and F_{conv} is identically zero.
19. Exact-adiabat test: arbitrary log-pressure grids give zero flux to roundoff/discretisation tolerance.
20. Manufactured one-interface case: w , ell , F_{conv} and K_{zz} match a hand calculation.
21. Column enthalpy test: zero-flux boundaries conserve $\sum(cp T \Delta m)$.
22. Analytic equilibrium test: final $T(P)$ has slope $2/7$ and the normalization fixed by initial enthalpy.

- 23. Resolution, alpha and timestep sweeps; transient forward-Euler error should approach first order in Δt while the converged state is Δt -independent below the stability limit.
- 24. Near-adiabat overshoot test: no layer freezes spuriously subadiabatic because of an excessive step.

EXIT GATE All converged cases reach the same analytic adiabat within declared tolerances; alpha changes relaxation rate but not equilibrium; enthalpy drift and boundary-flux error remain below tolerance; stable/unstable timestep boundary is documented.

2. Realistic H_2 thermodynamics and hydrostatics

Objective. Remove the constant-cp simplification without adding radiation or chemistry.

Build. Temperature-dependent cp and κ_{ad} ; consistent enthalpy function $h(T)$; optional H_2/He mixture; recomputed height, density, and gravity profile.

Variables to sweep. Temperature range, pressure range, H_2/He fraction, constant versus varying gravity, grid placement.

- 25. Thermodynamic identity tests: $dh/dT = cp$ and EOS consistency.
- 26. Numerically integrated isentrope benchmark replacing the power-law $2/7$ adiabat.
- 27. Conservation test in enthalpy rather than cp T.
- 28. Round-trip hydrostatic test: pressure-to-height and height-to-pressure remain consistent under grid refinement.

EXIT GATE The solver converges to a grid-independent variable-thermodynamic adiabat and conserves the correct total enthalpy.

3. Staggered absorption-only radiative transfer

Objective. Build the simplest independently testable radiative operator before coupling it to MLT or production opacities.

Build. Cell-centred T, P and opacity; $N+1$ upward/downward fluxes; no scattering; layer optical depth from $\kappa_{\text{ad}} \Delta P/g$; explicit top and lower boundary conditions. Derive the layered two-stream equations as a tridiagonal system.

Variables to sweep. N, optical-depth spacing, angular/diffusivity factor, spectral points or grey bands, top irradiation, lower flux/temperature, numerical precision.

- 29. Transparent, isothermal, single-layer, pure-absorption and optically thick limiting cases.
- 30. Thomas solve versus a dense linear solve for small columns.
- 31. Tridiagonal solution versus independent upward/downward sweeps where the absorption-only equations decouple.

- 32. Conservative layer heating from interface-flux divergence and exact accounting of top/bottom boundary flux.
- 33. Grid and optical-depth convergence; positive intensity/flux and monotonic attenuation tests.
- 34. NumPy reference versus jitted and batched JAX implementation in float64.

EXIT GATE All three reference routes agree within tolerance; analytic limits and column energy close; the JAX result is numerically equivalent after compilation.

4. Fixed-composition H₂ radiative–convective equilibrium

Objective. Combine the validated MLT and absorption-only radiative operators on one staggered finite-volume grid.

Build. Fixed grey or prescribed spectral opacity, shared interface-flux convention, explicit operator-split reference integration, HELIOS adapter/parity cases, and existing HELIOS convective adjustment retained as a benchmark.

Variables to sweep. Layer count and pressure range, irradiation/internal flux, optical-depth profile, lower boundary, alpha, radiative/convective update cadence, operator tolerances.

- 35. Manufactured prescribed-heating equilibrium before activating the full radiative solve.
- 36. Constant $F_{\text{rad}} + F_{\text{conv}}$ and grid convergence of $T(P)$, RCB pressure, and flux partition.
- 37. MLT versus HELIOS convective adjustment in the efficient-convection limit.
- 38. Convection-off regression and array-orientation/sign tests at the HELIOS boundary.
- 39. Frozen-operator experiments that separately measure stable dt_{rad} and dt_{MLT} , physical relaxation, call counts, and wall time.
- 40. Restart and hot/cold initial-profile convergence tests.

EXIT GATE Fixed-composition H₂ reaches the same RCE under timestep refinement and independent solver paths; total flux closes; the radiative versus convective bottleneck is measured rather than assumed.

5. Production gas-opacity infrastructure

Objective. Replace toy opacity with reproducible species tables and interpolation while keeping composition fixed.

Build. A project-owned opacity-provider interface; shone/DACE download and cache patterns where suitable; P–T–wavelength or correlated-k interpolation; mixture rules; units, provenance, bounds, and cache invalidation.

Variables to sweep. Species list, line/continuum source, spectral grid, P–T bounds, abundance basis, interpolation order, extrapolation policy, precision, correlated-k settings.

- 41. Exact-grid-node recovery and synthetic analytic interpolation fixtures.
- 42. Unit conversions, species-name mapping, abundance-weighting, and zero-opacity limits.
- 43. Out-of-domain and cache-invalidation tests with explicit failures rather than silent extrapolation.
- 44. Opacity-source comparison at selected P–T points and reproducible spectra/heating rates.
- 45. Gradient checks where JAX automatic differentiation will later support RCE Jacobians.
- 46. Benchmark interpolation and transfer separately over layers, wavelengths and g-points.

EXIT GATE A fixed-composition opacity case is reproducible from recorded provenance, passes interpolation/units tests, and produces grid-converged fluxes without chemistry feedback.

6. Fast steady RCE solver

Objective. Reduce wall-clock cost without changing the validated fixed-composition equilibrium.

Build. JAX jit/vmap over spectral samples; batched tridiagonal solve; semi-implicit MLT or direct total-flux residual solve; line search/trust region; explicit NumPy reference retained.

Variables to sweep. Solver tolerances, Jacobian/automatic-differentiation strategy, update clipping, continuation schedule, opacity-refresh threshold, hardware and batch size.

- 47. Solution equivalence against the explicit reference across the RCE benchmark suite.
- 48. Thomas versus JAX tridiagonal parity and compilation-versus-steady-runtime accounting.
- 49. Residual monotonicity and recovery from poor initial profiles.
- 50. Scaling with N, spectral resolution, g-points and optical-depth range.
- 51. No hidden dependence of the converged state on clipping, cache threshold, update ordering or solver path.

EXIT GATE The accelerated solver reproduces reference T(P), fluxes and RCB within tolerance and shows a documented speedup in the workload regime where JAX batching is advantageous.

7. One-way VULCAN coupling

Objective. Use the fixed-composition RCE state to drive photochemistry without allowing chemistry to alter thermodynamics or opacity.

Build. Interpolation of T(P), interface Kzz, pressure, gravity and composition metadata to the VULCAN grid; explicit background/overshoot Kzz policy.

Variables to sweep. VULCAN resolution, network, stellar UV, boundary conditions, Kzz floor/overshoot, chemical tolerances and opacity-composition freeze policy.

- 52. Conservative/shape-preserving interpolation and unit checks ($\text{m}^2 \text{s}^{-1}$ versus $\text{cm}^2 \text{s}^{-1}$).
- 53. Species/element conservation and chemical steady-state diagnostics.

- 54. Kzz sensitivity and quench-level tests.
- 55. Comparison with a prescribed-Kzz VULCAN baseline.
- 56. Audit that VULCAN output does not feed back into Stage 7 radiation.

EXIT GATE VULCAN converges reproducibly from RCE output; T and Kzz mapping is traceable; quench behaviour is grid stable; the coupling remains demonstrably one way.

8. Dry rocky-planet atmospheres and a surface

Objective. Move to the primary science target while composition and opacity remain prescribed.

Build. General multi-component EOS and cp; surface energy balance; sensible heat; surface-limited mixing length; fixed-composition gas opacity; optional Ledoux diagnostic without changing the Schwarzschild baseline.

Variables to sweep. N₂, CO₂, H₂O-as-noncondensing-tracer and mixtures; surface T/albedo/emissivity; pressure, gravity, stellar spectrum, internal flux and sensible-exchange coefficient.

- 57. Dry-adiabat benchmarks for pure gases and selected mixtures.
- 58. Surface plus atmosphere energy-budget closure.
- 59. AGNI intercomparison for matched dry fixed-composition cases.
- 60. Sensitivity of near-surface flux to mixing-length limiter and sensible exchange.
- 61. H₂ and absorption-only radiative regressions remain unchanged.

EXIT GATE Matched dry rocky cases agree with independent calculations within declared tolerances; surface and column energy close; earlier gates remain green.

9. Phase change, rainout and latent energy with frozen opacity

Objective. Validate moist-process mass and energy bookkeeping before closing the composition–opacity loop.

Build. Saturation adjustment/rainout, gas–condensate–rained reservoirs, latent heating/transport as a distinct energy term, re-evaporation policy and composition-dependent buoyancy. Gas opacity is prescribed or deliberately frozen; cloud opacity is off.

Variables to sweep. Condensable identity/abundance, saturation curve, latent heat, precipitation efficiency, retention, re-evaporation and surface reservoir policy.

- 62. Single-condensable pseudo-moist adiabat benchmark.
- 63. Closed-box conservation of gas + condensate + rained-out mass and each relevant element.
- 64. Energy conservation including sensible and latent enthalpy.
- 65. Dry limit as condensable abundance tends to zero and latent heat tends to zero.

- 66. Comparison of VULCAN rainout-only behaviour with the energy-closed operator.
- 67. Frozen-opacity assertion prevents phase-change composition from silently changing radiation.

EXIT GATE Mass and total energy close across phase change; dry and pseudo-moist limits are recovered; results are grid converged independently of interactive opacity.

10. Two-way composition–thermodynamic–gas-opacity feedback

Objective. Close the VULCAN feedback loop only after radiation, opacity interpolation, surfaces and phase-change bookkeeping are independently trusted.

Build. Outer fixed-point or block iteration; updated μ , cp , ∇_{ad} and gas opacity from VULCAN composition; under-relaxation or Anderson acceleration; selective opacity refresh with explicit error control.

Variables to sweep. Coupling relaxation, update frequency, chemistry tolerance, opacity threshold, convergence norms, initial composition and condensation-coupling mode.

- 68. Joint residuals for flux, temperature, abundances, elemental inventory, thermodynamics and opacity.
- 69. Path-independence from hot/cold and equilibrium/disequilibrium initial states.
- 70. Freeze-one-block experiments isolating radiation, convection, chemistry, thermodynamics and opacity feedback.
- 71. Restart tests and cache-threshold refinement.
- 72. Turnover, mixing, radiative and chemical times remain distinct from outer iteration count.

EXIT GATE All blocks converge simultaneously; restarting changes no field above tolerance; energy and elemental budgets close; the frozen-opacity limit reproduces Stage 7 or 9 as appropriate.

11. Fully coupled condensing rocky RCE

Objective. Combine the validated rocky surface, phase-change energy operator and interactive gas-opacity feedback.

Build. Joint solution of surface temperature, atmospheric RCE, chemistry/transport, condensation/rainout reservoirs and gas opacity. Continue to exclude scattering/cloud opacity so the new feedback is isolated.

Variables to sweep. Condensable inventory, surface reservoir, rainout/re-evaporation, chemistry cadence, opacity cadence, coupling acceleration and convergence tolerances.

- 73. Grid-independent moist RCE and surface/atmosphere total-energy closure.
- 74. Path independence and restart reproducibility.
- 75. Dry, frozen-opacity and no-chemistry limits recover Stages 8–10.

76. Matched AGNI cases where assumptions overlap.

77. Process-tendency and wall-time profiles identify the actual coupled bottleneck.

EXIT GATE The condensing rocky solution closes mass, elemental and energy budgets and is insensitive to numerical path within declared tolerances.

12. Scattering, clouds and advanced physical-time experiments

Objective. Add radiative and dynamical complexity only after absorption-only coupled solutions are established.

Build. Two-stream scattering, Rayleigh scattering, condensate/cloud opacity and later microphysics; Ledoux stability, overshoot, non-ideal EOS, dissociation/ionisation and truly time-dependent perturbations as science requires.

Variables to sweep. Single-scattering albedo, asymmetry factor, cloud particles, molecular-weight gradient, overshoot decay, forcing amplitude/period and model-dependent transport parameters.

78. Pure-scattering, absorbing-scattering and zero-scattering limits.

79. Comparison with HELIOS, DISORT or another trusted solver for matched columns.

80. Each added feature has a zero-effect limit reproducing the previous stage.

81. Time-dependent experiments compare response functions and physical relaxation times, not solver iteration counts.

82. Benchmark against multidimensional or published results where analytic solutions do not exist.

EXIT GATE Every extension has a feature-specific validation dossier before production use; scattering/cloud-off runs reproduce Stage 11.

5. Verification and validation programme

Testing should be hierarchical. A high-level agreement cannot compensate for a failed conservation or sign test.

Level	Purpose	Examples
Unit	Formula, shape, sign, units	Hp, cp, gradients, optical depth, interface interpolation, Kzz/opacity units
Invariant	Quantities that must not drift	Enthalpy, total energy, elemental abundance, zero boundary flux
Manufactured	Known local or global solution	One-interface MLT flux; single-layer transfer; imposed flux divergence
Convergence	Numerical correctness	N, dt, nonlinear tolerance, grid type, interpolation order
Regression	Prevent accidental behavioural change	Stored benchmark summaries, not giant opaque output files

Level	Purpose	Examples
Intercomparison	Independent implementation check	Dense/Thomas/sweep transfer; HELIOS adjustment; AGNI; Exo-FMS/Lee; GENESIS
Science benchmark	End-to-end physical credibility	H ₂ RCE; dry CO ₂ /N ₂ rocky case; latent-energy case; chemical quench case

Stage 1 analytic solution

For constant c_p and $nabla_{ad}$, the final connected convective region is

$$T_i = A (P_i/P_0)^{\nabla_{ad}}, \quad A = [\sum_i c_p T_{i,0} \Delta m_i] / [\sum_i c_p (P_i/P_0)^{\nabla_{ad}} \Delta m_i].$$

Evaluate both slope and normalization. A profile can have the right gradient while carrying the wrong total enthalpy.

Baseline experiment matrix

Axis	Baseline values	What it proves
Resolution N	25, 50, 100, 200, 400	Grid convergence and computational scaling
Mixing length α	0.25, 0.5, 1, 2, 4	Same closed-column equilibrium; different relaxation rates
Time-step factor	0.02, 0.05, 0.1, 0.2, 0.5, 1, $2 \times$ stability estimate	Stable range, first-order transient convergence, failure boundary
Grid	Uniform log P; smoothly perturbed log P	No dependence on a special regular grid
Initial profile	Power-law superadiabat; local unstable layer; multi-zone; exact adiabat; stable profile	Correct mixing region and limiting behaviour
Pressure range	At least two ranges spanning different cell masses	Robust pressure-coordinate implementation

Acceptance metrics

Metric	Definition	Initial target / interpretation
Flux residual	$\max F_{\text{bottom}} - F_{\text{top}} $ or equilibrium residual	Set relative to imposed/characteristic flux; report absolute and relative
Enthalpy drift	$ H(t) - H(0) - \text{boundary input} / H(0)$	Aim $\leq 1e-10$ in closed reference runs; relax only with documented solver limits
Adiabat slope error	RMS and max of $nabla - \nabla_{ad}$ in mixed region	Must decrease with N and solver tolerance
Temperature error	RMS and max relative to analytic $T(P)$	Must include normalization, not slope alone

Metric	Definition	Initial target / interpretation
Potential-temperature/entropy spread	Relative spread within a mixed region	Independent confirmation of adiabatic homogenisation
Boundary error	Fconv at closed top and bottom	Exactly zero by construction
Timestep order	Error ratio under dt, dt/2, dt/4	Approximately first order for forward Euler transient
Grid order	Observed error scaling with N	Measure; do not assume theoretical order at moving RCBs
Performance	Wall time and number of radiation/chemistry/MLT calls	Report separately from physical simulated time

TOLERANCE POLICY The numerical values above are starting targets, not scientific constants. Lock final tolerances only after double-precision baselines reveal roundoff, discretisation and solver floors. Every relaxed tolerance needs a reason and an error budget.

Regression policy

83. Store compact reference summaries: norms, budgets, RCB pressure, selected layer values, and metadata.
84. Version benchmark data with the equation/closure version. A physics change may intentionally update a baseline; a refactor may not.
85. Run fast unit/invariant tests on every commit; run N–alpha–dt sweeps on scheduled CI or before release.
86. Treat warnings, non-convergence, NaNs, negative temperatures/densities, and unit mismatches as test failures.

6. Variable hierarchy and experiment design

Label every varied quantity as numerical, closure, boundary, planetary, radiative, chemical, or condensate. This prevents grid/timestep sensitivity from being misread as physical uncertainty.

Class	Variables	Treatment
Numerical	N, pressure grid, dt/CFL factor, nonlinear tolerance, interpolation, update cadence	Converge away; never calibrate to observations
MLT closure	alpha, prefactor, length limiter, overshoot law, Kzz floor	Sensitivity/calibration parameters with explicit metadata
Thermodynamic	cp, mu, nabl_a_d, EOS, enthalpy, dissociation	Physical inputs with unit/identity tests
Planet/boundary	g, radius, pressure range, surface T/albedo/emissivity, internal flux	Science scenario inputs

Class	Variables	Treatment
Radiative	two-stream closure, angular factor, stellar spectrum, boundaries, opacity, scattering, spectral/g-point resolution	Benchmark and convergence dimensions
Opacity/data	species source, P–T grid, interpolation, mixture rule, extrapolation, cache tolerance, precision	Provenance, unit and numerical-error dimensions
Chemical	network, rate data, UV, boundary fluxes, initial abundances, Kzz	Conservation and stiffness dimensions
Condensate/cloud	saturation, latent heat, rainout, particles, opacity	Add only after mass/energy closure

Recommended sensitivity strategy

87. One-factor sweeps for Stage 1 to identify implementation errors and stability thresholds.
88. Small factorial tests for interactions among N, alpha, and timestep factor.
89. Matched-case intercomparisons at a few physically interpretable points, not only large random grids.
90. Latin-hypercube or Sobol sampling only after the solver is reliable and the goal is physical sensitivity rather than debugging.
91. Emulators/surrogates only for repeated science grids; never use them as the validation reference.

7. Timescale framework

CRITICAL DISTINCTION Numerical timestep, physical process timescale, and wall-clock runtime are different quantities. Report all three with different names and units.

Quantity	Definition / measurement	Use
Numerical dt_MLT	Stability- and accuracy-limited update step	Integrator control only
Turnover time	$\tau_{\text{turn}} = \ell / w$	Characteristic eddy travel time
Mixing time	$\tau_{\text{mix}} = L_{\text{mix}}^2 / K_{\text{zz}}$; often H_p^2 / K_{zz}	Quenching and tracer transport comparison
Radiative time	Perturbation response $\delta T / \delta (dT/dt)_{\text{rad}}$; later radiative Jacobian eigenmodes	Thermal response and multirate design
Chemical time	Reaction/Jacobian eigenvalues or controlled species perturbation	Quenching and stiffness diagnosis
Coupling convergence	Outer iterations or block residual reduction	Numerical algorithm performance
Wall-clock time	Measured CPU/GPU seconds and call counts	Computational cost

For the explicit MLT reference solver, begin with a diffusion-like stability estimate and a fractional-temperature limiter:

$$dt_MLT \leq C \min_i (\Delta z_i^2 / K_{H,i}), \text{ with } K_H \sim w \text{ ell; additionally enforce } \max_i |\Delta T_i| / T_i < \epsilon_{T_T}.$$

Use one global substep for neighbouring layers so a single interface flux is applied consistently to both adjacent control volumes. Local asynchronous timesteps require a deliberately conservative algorithm and are not part of Stage 1.

Process-timescale experiments

92. Convection: perturb entropy locally in a convective column and fit the decay of modes; compare with ℓ/w and H_p^2/K_{zz} .
93. Radiation: perturb one layer and broader vertical modes, recompute two-stream/HELIOS heating, and estimate the response matrix.
94. Chemistry: perturb selected species around steady state and fit relaxation; compare with chemical Jacobian eigenvalues.
95. Coupled system: perturb T or composition after Stage 10 and track cross-response without changing numerical tolerance.
96. Quenching: locate where $\tau_{\text{chem}}(P)$ and $\tau_{\text{mix}}(P)$ cross, then verify against the full kinetic abundance profile.

Radiative-versus-convective bottleneck experiment

At Stage 4, report three separate answers: (1) which operator sets the smallest stable explicit substep, (2) which physical mode has the longest relaxation time, and (3) which operator consumes the most wall-clock time. Run radiation-only, convection-only and coupled cases from the same perturbed state. Log dt_{rad} , dt_{MLT} , residual reduction per call, number of calls, compilation time, steady execution time and final solution error. The same operator need not dominate all three measures.

8. Increasing calculation rate

Optimisation should preserve the validated operator. Introduce accelerations in the following order, measuring solution error and wall time at each step.

Tier	Method	Expected value	Primary risk / required check
A. Reference	Vectorised NumPy; Thomas + sweep/dense transfer references; explicit global dt	Transparent baseline and profiling	Small stable dt; verify call-level conservation
B. Spectral	JAX jit/vmap; batched tridiagonal solves over wavelengths/g-points; float64 parity	Compiled high-throughput radiation	Compilation overhead, batch size and precision

Tier	Method	Expected value	Primary risk / required check
C. Safe local	Adaptive global dt, event-aware stopping, reuse opacity when state change is bounded	Fewer unnecessary steps/evaluations	Cache invalidation and tolerance dependence
D. Convective	Semi-implicit nonlinear diffusion or backward Euler with Newton/Picard	Remove explicit MLT stability bottleneck	Reproduce explicit equilibrium and conserve flux
E. Full RCE	Direct residual solve with Newton/LM, line search, clipping and continuation	Avoid slow evolution through long radiative times	Non-convexity, initial profiles, Jacobian cost
F. Jacobian	Structured analytic or JAX automatic derivatives; JFNK/preconditioning	Better scaling with N and variables	Derivative correctness and memory
G. Coupling	Block Gauss-Seidel, under-relaxation, Anderson acceleration, selective opacity refresh	Fewer outer iterations	False convergence of one block
H. Hardware	Fuse flux/divergence kernels; parallel spectral and parameter sweeps	High throughput after validation	Transfer overhead and backend divergence
I. Surrogates	Reduced chemistry or emulators for repeated science grids	Production-scale studies	Validity domain and conservation loss

Recommended steady-solver path

97. Retain forward Euler as the truth-maintaining transient reference.
98. Retain NumPy Thomas, dense and directional-sweep radiative solutions as small-column references; compare every JAX kernel against them.
99. Implement a conservative backward-Euler or semi-implicit convection-only step and compare against the reference.
100. Form a steady residual from interface total-flux differences, following AGNI's energy-conservation framing.
101. Use line search and update clipping; apply continuation from grey → fixed production opacity → variable gas composition → phase change → scattering/cloud opacity.
102. Exploit the local/banded structure of MLT and the nonlocal structure of radiation explicitly in the preconditioner.
103. Profile before porting the new operator to the GPU; radiative transfer may remain the dominant cost.

Performance benchmark record

Every benchmark should log N, spectral points/bands, chemical species/reactions, hardware, precision, solver, iterations, rejected steps, radiation calls, chemistry calls, MLT calls, wall time, peak memory, final residuals, and solution error versus the reference.

9. HELIOS–VULCAN coupling protocol

Stage 7 is deliberately one way; Stage 10 activates sequential fixed-point feedback. A monolithic radiation–convection–chemistry Jacobian is a later optimisation, not the first integration target. Phase-change and surface modules are validated before the two-way gas-opacity loop, even though fully coupled condensing RCE is completed only after that loop exists.

104. The fixed-composition RCE solver (project two-stream kernel and/or HELIOS comparison path) outputs centre $T(P)$, thermodynamics, interface w , ℓ , F_{conv} and $K_{\text{zz_conv}}$.
105. The adapter constructs total K_{zz} using the declared convective, overshoot, wave/background and floor components, then interpolates T and K_{zz} to VULCAN.
106. In Stage 7, VULCAN integrates chemistry/photochemistry and returns diagnostic composition, but the RCE opacity and thermodynamics remain frozen.
107. In Stage 10, a property layer computes updated μ , c_p , ∇_{ad} , EOS quantities and gas-opacity inputs from VULCAN composition on the radiative grid.
108. The RCE solver recomputes temperature and mixing. Under-relax temperature, composition or gas opacity if the outer loop oscillates.
109. Stage 11 additionally activates the already validated phase-change reservoirs and latent-energy term; Stage 12 adds condensate/cloud opacity and scattering.
110. Stop only when flux residuals, temperature changes, abundance changes, elemental budgets and opacity changes all satisfy their own criteria.

Exchange field	Native location	Coupling rule
T , P , g , ρ , μ , c_p , ∇_{ad}	Layer centres	Shape-preserving interpolation in $\log P$ where appropriate
F_{conv} , w , ℓ , K_{zz}	Interfaces	Preserve interface meaning; convert $\text{m}^2 \text{s}^{-1} \leftrightarrow \text{cm}^2 \text{s}^{-1}$ explicitly
Mixing ratios/number densities	VULCAN centres	Conservative/positive mapping; renormalise only with an auditable rule
Gas-opacity inputs	Radiative layer/species representation	Frozen in Stages 7/9; interactive from Stage 10; cache with an audited threshold
Condensed/rained mass	Layer + reservoir	Budgeted in Stage 9; coupled to RCE in Stage 11; cloud opacity deferred to Stage 12

Kzz policy

Publish each component separately: $K_{\text{zz_conv}}$ from MLT, $K_{\text{zz_overshoot}}$, $K_{\text{zz_background}}$, and $K_{\text{zz_total}}$ used by VULCAN. This avoids interpreting a numerical floor or radiative-zone prescription as a direct prediction of convection.

10. Relationship to existing codes and studies

Reference	Similarity / lesson	Project use
AGNI	N pressure centres, N+1 flux interfaces; MLT Fconv; Kzz; separate latent/sensible terms; nonlinear energy-flux residual solve; rocky-planet surface treatment	Closest architecture. Adopt staggering, modular flux sum, surface-aware mixing length and steady-residual concept; use matched dry rocky cases for intercomparison.
Current HELIOS	Same centre/interface grid concept and GPU radiative transfer; existing enthalpy-conserving dry convective adjustment	Integration/parity target and efficient-convection benchmark. Keep adjustment and radiation as selectable comparison paths while the project-owned kernels mature.
shone	JAX-based opacity download/cache and P–T–wavelength interpolation patterns; transmission tools	Strong starting point for opacity data plumbing and interpolation. Keep the project-owned emission/RCE kernel independently tested; do not depend on an unvalidated draft emission implementation.
JAX	Compiled array execution, vmap batching, automatic differentiation and a batched tridiagonal solve	Production framework for spectral/g-point batches and later Jacobians. Retain NumPy Thomas/dense references and benchmark compilation overhead.
VULCAN	1D kinetics/photochemistry with transport, condensation/rainout options, T(P) and Kzz inputs	Chemistry engine. Couple first one-way, then return composition-dependent opacity and thermodynamics.
Lee, Tan & Tsai / Exo-FMS	Pressure-coordinate time-dependent MLT, Kzz = w ell, subcycling; explicit steps around or below 1 s in reported tests	Guide explicit implementation, stability tests, subcycling, Kzz and overshoot diagnostics. Note that the published application is a 3D GCM, not the final 1D solver.
Robinson & Marley	1D time-stepping radiative–convective model; mixing-length sensitivity and thermal response times	Reference for transient convergence, timestep behaviour and physical perturbation experiments.
GENESIS	MLT embedded in a Rybicki/Newton-like radiative–convective solve; parcel optical-depth/radiative exchange in the closure	Benchmark fuller MLT near inefficient convection/RCB. Defer this closure until the simple model is validated.
Joyce & Tayar	Modern review of assumptions, free parameters and limitations of 1D MLT	Theory and notation reference; use to document calibration limits.

The comparison reflects the cited papers, public repositories and documentation available on 8 August 2026. shone’s public opacity tools are treated as reusable infrastructure; any emission/scattering code must pass this project’s independent validation before adoption. Verify version-specific details before freezing dependencies.

What to adopt now, benchmark, and defer

Adopt now	Benchmark explicitly	Defer until justified
Interface flux staggering; conservative divergence; simple MLT; absorption-only two stream; NumPy/JAX parity; separate Kzz components	Thomas/dense/sweep transfer; HELIOS adjustment/radiation; AGNI dry cases; analytic H ₂ adiabat; Lee/Robinson transients	Scattering/cloud opacity; parcel radiative-loss closure; Ledoux/overshoot; monolithic coupled Jacobian; cloud microphysics

11. Project governance, deliverables and release gates

Use work packages with evidence-based release gates rather than date-only milestones. Calendar estimates depend strongly on available developer time, opacity-data access and representative CPU/GPU hardware.

Release	Minimum deliverables	Release evidence
R0 — Reference kernel	Package, docs, Stage 0–1 tests, reproducible sweep runner	Closed-H ₂ validation report and CI
R1 — Radiative kernel	Staggered absorption-only two stream; NumPy Thomas/dense/sweep references; JAX parity	Analytic limits, flux conservation and solver-equivalence report
R2 — Fixed H ₂ RCE	MLT + radiation, HELIOS parity/adaptor, fixed opacity and bottleneck instrumentation	Flux/RCB convergence and dt/relaxation/wall-time report
R3 — Opacity + one-way chemistry	Production gas-opacity provider, shone adaptor where suitable, T/Kzz export to VULCAN	Opacity provenance/interpolation tests and quench baseline
R4 — Dry rocky surface	General EOS, fixed-composition opacity, surface and sensible heat	AGNI/matched-case comparison and surface energy closure
R5 — Phase change module	Rainout + latent energy + reservoirs with frozen/prescribed opacity	Mass/energy closure and dry/pseudo-moist limits
R6 — Two-way gas feedback	Outer composition–thermodynamic–gas-opacity coupling and acceleration	Joint convergence, restart/path and frozen-opacity-limit tests
R7 — Condensing rocky RCE	Surface + phase change + chemistry + interactive gas opacity	Full mass/element/energy closure and grid-independent moist RCE
R8 — Scattering and clouds	Scattering solver, condensate opacity and later microphysics	Scattering limits and trusted-solver intercomparison

Definition of done for each work package

- 111. Equations and sign/unit conventions documented.
- 112. Unit, invariant, convergence and regression tests pass.
- 113. A reproducible experiment configuration and compact results file are committed.
- 114. Failure modes and valid parameter domain are documented.
- 115. Performance is measured on at least one named reference machine.
- 116. Any change to a physics baseline has a short decision record.

Risk register

Risk	Early warning	Mitigation
Explicit MLT is prohibitively stiff	dt shrinks rapidly with N or alpha	Keep reference solver; add conservative semi-implicit step before full HELIOS production

Risk	Early warning	Mitigation
Near-adiabat freezing/overshoot	Patchy stable interfaces and entropy noise	Global dt, fractional-change limiter, monotone/nonlinear solve, entropy diagnostics
Radiative discretisation is wrong but converges	Thomas result disagrees with dense/sweep or analytic limits	Maintain three independent small-column routes and boundary-flux invariants
Opacity interpolation is silently invalid	Out-of-range P–T states or discontinuous heating	Record provenance/domain; fail loudly; use synthetic fixtures and cache-threshold refinement
JAX changes precision or hides overhead	CPU/JAX parity loss or compilation dominates runtime	Enable/test float64; separate compile and execute timings; retain NumPy reference
Radiation–chemistry oscillation	Outer residual alternates or opacity jumps	Block under-relaxation, Anderson acceleration, continuation and selective updates
Kzz overinterpretation	Radiative-zone mixing attributed to convection	Output Kzz components separately and document floors/overshoot
Condensation breaks energy closure	T changes when mass is removed without latent term	Keep rainout and energy terms explicit; maintain reservoir budgets
GPU port obscures bugs	CPU/GPU divergence or nondeterministic residuals	CPU reference and small deterministic GPU parity tests
Scope expands before validation	Many features, no analytic benchmark	Enforce stage gates and feature flags

12. Immediate implementation backlog

The first development cycle should end with Stage 1 evidence, not a Guillot-profile production run.

117. Freeze existing v2/v3 code under legacy/ and record one reproducible legacy baseline.
118. Create grid, constant-H₂ thermodynamics, MLT closure, energy operator, diagnostics and solver modules.
119. Correct $cp = (7/2)R/\mu$ for the baseline and use enthalpy consistently.
120. Reverse the staggering to centre T / interface Fconv and define upward-positive flux.
121. Implement zero-flux boundaries and one global explicit substep.
122. Implement analytic enthalpy-normalised final adiabat.
123. Write stable, exact-adiabat, manufactured-flux, sign and enthalpy tests.
124. Build the N–alpha–dt–grid–profile sweep runner with deterministic metadata.
125. Generate convergence plots/tables and identify stable dt limits.
126. Only after the exit gate, specify the Stage 3 staggered absorption-only kernel, boundary conditions, Thomas/dense/sweep references and JAX parity tests.

FIRST MILESTONE For a closed convection-only constant- c_p H_2 column, every tested resolution and every non-zero α must converge to the same enthalpy-conserving analytic adiabat. Timestep refinement may change the transient path, but not the final state.

13. References and code resources

- [R1] Nicholls, H., Pierrehumbert, R. & Lichtenberg, T. (2025). AGNI: A radiative-convective model for lava planet atmospheres. *Journal of Open Source Software* 10, 7726. [Open reference](#)
- [R2] AGNI model description and nonlinear energy-residual solver documentation (current documentation accessed 8 August 2026). [Open reference](#)
- [R3] AGNI atmospheric convection and Kzz documentation (current documentation accessed 8 August 2026). [Open reference](#)
- [R4] Nicholls, H. et al. (2025). Convective shutdown in the atmospheres of lava worlds. *MNRAS* 536, 2957–2971. [Open reference](#)
- [R5] Malik, M. et al. (2019). Self-luminous and irradiated exoplanetary atmospheres explored with HELIOS. *Astronomical Journal* 157, 170. [Open reference](#)
- [R6] HELIOS open-source repository. [Open reference](#)
- [R7] Tsai, S.-M. et al. (2017). VULCAN: an open-source, validated chemical kinetics Python code for exoplanetary atmospheres. *ApJS*; preprint and model description. [Open reference](#)
- [R8] Tsai, S.-M. et al. (2021). A comparative study of atmospheric chemistry with VULCAN. *ApJ* 923, 264. [Open reference](#)
- [R9] VULCAN open-source repository. [Open reference](#)
- [R10] Lee, E. K. H., Tan, X. & Tsai, S.-M. (2024). Dynamically coupled kinetic chemistry in brown dwarf atmospheres II. *MNRAS* 529, 2686–2701. [Open reference](#)
- [R11] Exo-FMS one-column correlated-k implementation, including MLT options. [Open reference](#)
- [R12] Robinson, T. D. & Marley, M. S. (2014). Temperature fluctuations as a source of brown dwarf variability. *ApJ* 785, 158. [Open reference](#)
- [R13] Marley, M. S. & Robinson, T. D. (2015). On the cool side: modeling the atmospheres of brown dwarfs and giant planets. *ARA&A* 53, 279–323. [Open reference](#)
- [R14] Gandhi, S. & Madhusudhan, N. (2017). GENESIS: new self-consistent models of exoplanetary spectra. *MNRAS* 472, 2334–2355. [Open reference](#)
- [R15] Joyce, M. & Tayar, J. (2023). A review of the mixing length theory of convection in 1D stellar modeling. *Galaxies* 11, 75. [Open reference](#)

[R16] ConvectionMLT project repository and current development baseline. [Open reference](#)

[R17] shone open-source repository: JAX-based opacity, chemistry and exoplanet-atmosphere tools. [Open reference](#)

[R18] shone opacity documentation and examples (current documentation accessed 8 August 2026). [Open reference](#)

[R19] JAX documentation: `jax.lax.linalg.tridiagonal_solve`. [Open reference](#)

[R20] JAX documentation: Just-in-time compilation, batching and execution model. [Open reference](#)

[R21] shone draft emission-spectrum pull request #19; treat as unvalidated until merged and independently tested. [Open reference](#)

Appendix A. Stage 1 run record

Every sweep result should include the following fields so numerical and physical comparisons remain reproducible.

Category	Required fields
Identity	git commit, code version, date, hostname/hardware, precision
Grid	N, Ptop, Pbottom, interface and centre construction, irregularity seed
Physics	mu, cp/EOS version, g, alpha, MLT prefactor, mixing-length rule, boundaries
Integrator	method, dt rule, safety factor, epsilon_T, max steps/time, tolerances
Initial condition	profile family, parameters, perturbation/mixed-region definition
Outcome	converged flag/reason, steps, simulated relaxation time, wall time
Budgets	initial/final enthalpy, drift, boundary flux integral
Errors	T RMS/max, slope RMS/max, entropy/potential-temperature spread, observed order
Diagnostics	min/max dt, max w, ell, Fconv, Kzz, overshoot count, stable/unstable interfaces

Appendix B. Decision points before physical expansion

Decision	Trigger / rule
Simple versus radiatively leaky MLT	Promote GENESIS/Henyey-like parcel radiative exchange only if matched tests show meaningful differences near optically thin/inefficient convection or the RCB.
Explicit versus steady solver	Retain explicit reference; use steady residual solve for production RCE after equivalence is demonstrated.
Radiative linear solver	Keep NumPy Thomas plus dense/sweep references. Use JAX tridiagonal solves for production only after float64 parity and workload-specific speedup are demonstrated.

Decision	Trigger / rule
Opacity dependency	Use shone-compatible opacity acquisition/interpolation where beneficial, behind a project-owned provider interface. Do not couple correctness to draft emission code.
Feedback ordering	Validate dry-surface and latent-energy modules with fixed/frozen opacity before two-way VULCAN gas-opacity feedback; complete condensing RCE only after the feedback loop is available.
Schwarzschild versus Ledoux	Use Schwarzschild for homogeneous dry stages. Add Ledoux when chemistry, condensation, or escape creates persistent μ gradients.
MLT Kzz outside convection	Do not extrapolate silently. Choose and label overshoot/background prescriptions per science case.
Moist treatment	Do not substitute <code>nabla_moist</code> into dry MLT as the final scheme without an energy budget. Add condensation, latent transport and reservoirs explicitly.
Time-dependent coupled physics	Attempt only after steady coupled solutions and separate process-response diagnostics are established.